Tesla Transformer and Partial Discharges

Demonstration and experiment in the h_da High Voltage Laboratory

EEN442 Erasmus lab week · Wednesday 21 October 2026

Lab leads: Prof. Dr.-Ing. Thomas Betz and Frank Radler

1. Purpose

Besides the breakdown experiments and the lightning impulse demonstration, the HV lab includes two short topics:

- **Tesla transformer:** how a spark gap and two coupled resonant circuits turn a 50 Hz supply into a high-frequency high voltage.
- **Partial discharges in solids:** how a sharp electrode inside solid insulation produces small discharges that do not bridge the electrodes but slowly destroy the material.

Both topics share one idea: a gap that breaks down every time the voltage across it reaches a fixed value produces a train of pulses whose number depends on the applied voltage.

2. The Tesla transformer

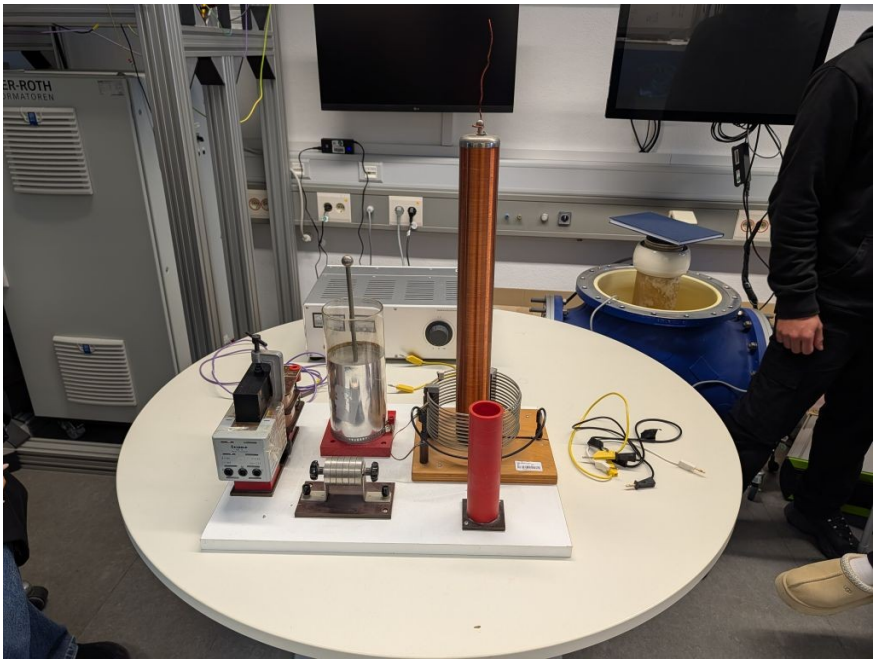


Figure 1. Spark-gap Tesla transformer on the demonstration table: secondary coil (tall copper tube), primary spiral at its base, capacitor and spark gap.

Circuit

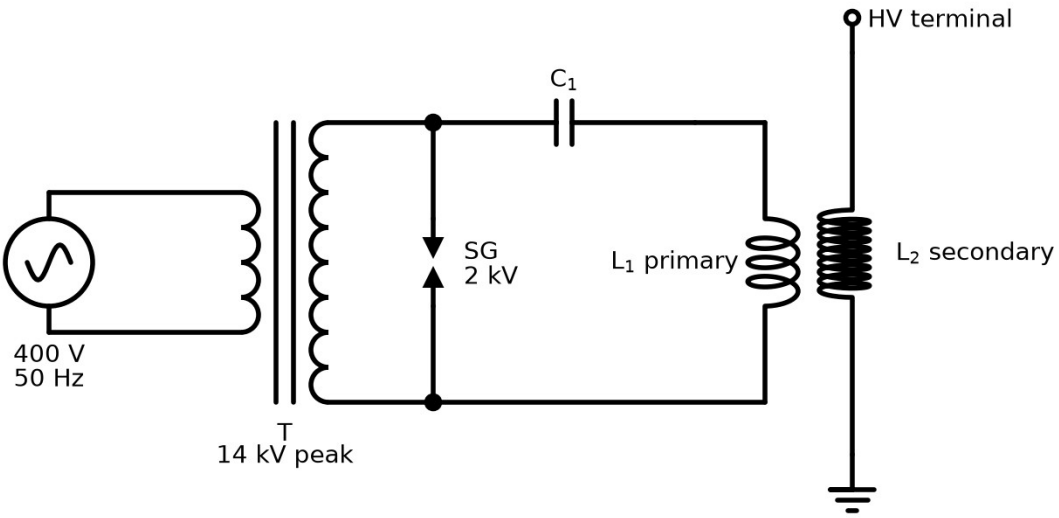


Figure 2. Circuit of the spark-gap Tesla transformer with the values used in the h_da demonstration.

A Tesla transformer is an air-core resonant transformer that produces a high voltage at low current and high frequency (Wikipedia, Tesla coil). In the h_da set-up:

- **Supply:** the 400 V mains supply feeds a high-voltage transformer T with a secondary peak of 14 kV.
- **Spark gap SG:** the gap breaks down at 2 kV and acts as a fast switch.
- **Primary circuit:** the capacitor C1 and the few-turn primary coil L1 form an LC circuit.
- **Secondary circuit:** the tall coil L2 with many turns of fine wire resonates with its own stray capacitance and the capacitance of the top terminal.

In the photo, the glass cylinder with a metal lining is probably the capacitor C1 (a Leyden jar) and the cylinder with stacked metal discs is probably the spark gap.

Operation cycle

1. The transformer T charges C1 until the voltage across the spark gap reaches 2 kV.
2. The gap fires and C1 discharges through L1. Current swings back and forth between C1 and L1 at the resonant frequency of the primary circuit.
3. The magnetic field of L1 induces an oscillating voltage in L2. Because L2 is tuned to the same frequency, energy builds up in the secondary over several cycles, as in two coupled pendulums.
4. When the energy has been passed on, the spark goes out, C1 recharges and the cycle starts again.

Both circuits must be tuned to the same resonant frequency:

$$f = 1/(2\pi\cdot\sqrt{(L1\cdot C1)}) = 1/(2\pi\cdot\sqrt{(L2\cdot C2)})$$

Typical Tesla transformers resonate between about 50 kHz and 1 MHz (Wikipedia, Tesla coil). Unlike a 50 Hz transformer, the output voltage is not set by the turns ratio but by the resonance and by how quickly the current changes, as noted on the whiteboard:

$$u = L\cdot di/dt$$

A fast change of current in a large inductance produces a high induced voltage. The whiteboard gives about 80 kV for the output of the demonstration coil.

How many sparks per cycle?

The spark gap fires every time the voltage across it has changed by 2 kV since the last spark. Over one 50 Hz cycle the supply voltage rises from 0 to 14 kV, falls to −14 kV and returns to 0: a total change of 4 × 14 kV = 56 kV. The number of sparks per cycle is therefore:

$$N = 4 \cdot \hat{U} / U_{gap} = 4 \cdot 14\ \text{kV} / 2\ \text{kV} = 28 \text{ sparks per } 20\ \text{ms}, \text{ or } 1400 \text{ sparks per second}$$

This is the "7 + 7 + 7 + 7 = 28 per 20 ms" on the whiteboard: 7 sparks in each quarter of the cycle. Figure 3 shows this idealised behaviour, which ignores the charging time of C1.

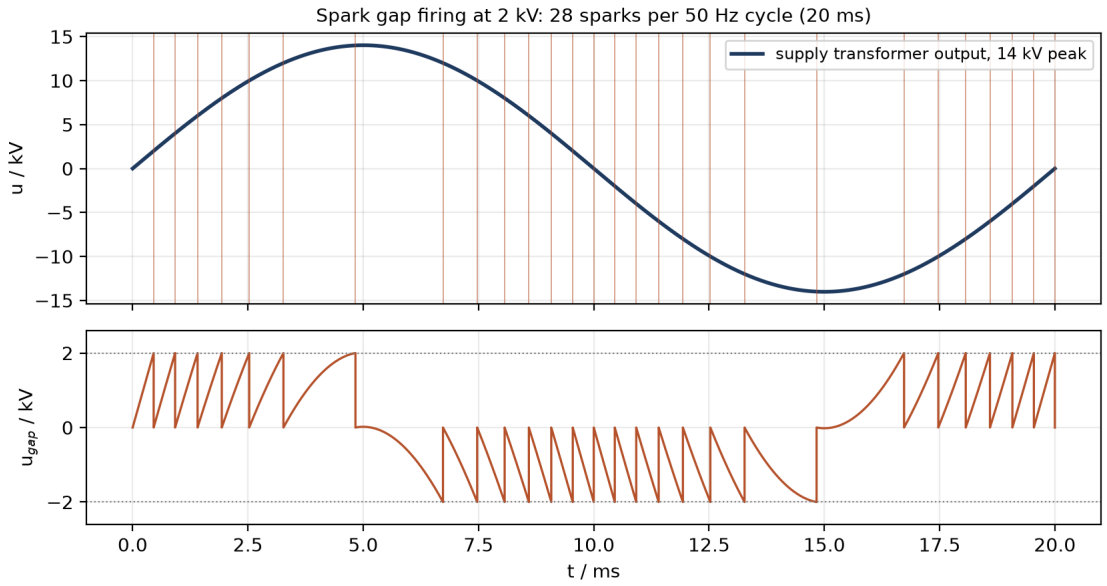


Figure 3. Idealised firing of a 2 kV spark gap on a 14 kV peak supply. Top: supply voltage with spark instants. Bottom: voltage across the gap.

Demonstrations

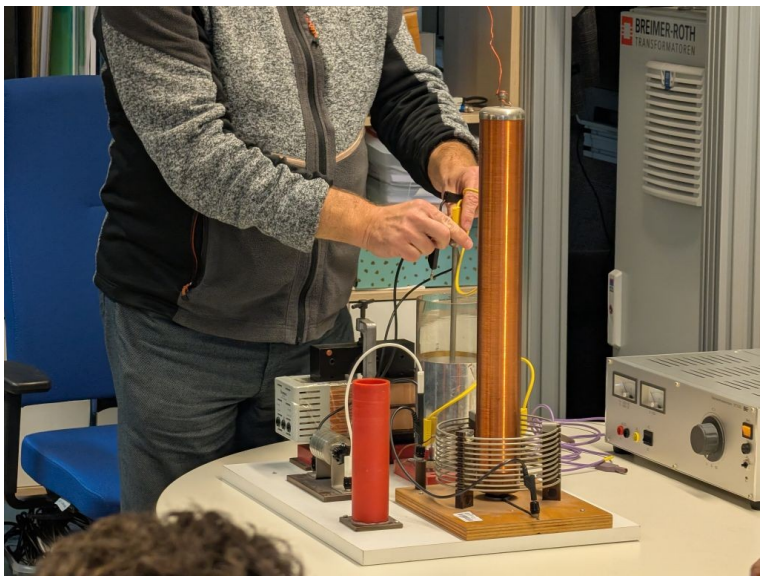


Figure 4. Connecting the spark-gap Tesla transformer.

At high frequency the output can light a fluorescent tube held nearby, with no wires connected, because the strong alternating field excites the gas in the tube (WWU Physics, High-Frequency Tesla Apparatus).

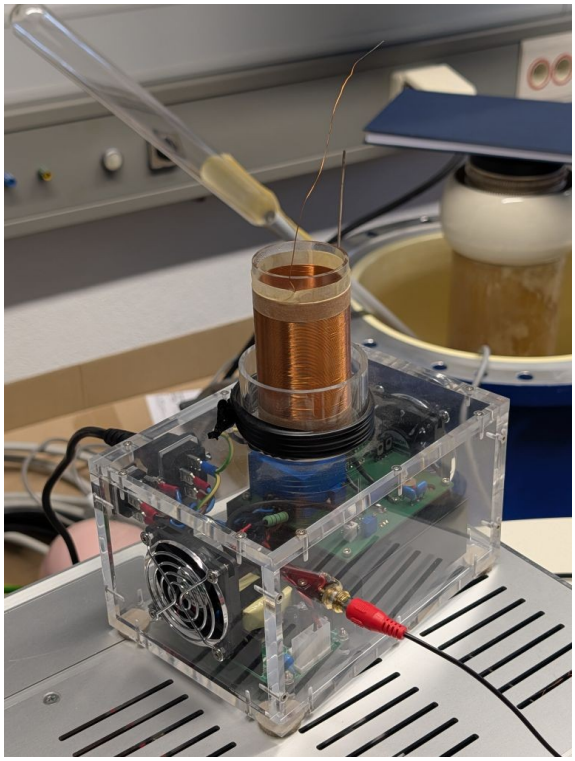


Figure 5. Small solid-state Tesla coil with a fluorescent tube held near the top.

The small coil in the acrylic box is a solid-state Tesla coil: electronics instead of a spark gap drive the primary. A music player is connected through the audio cable, so the spark is modulated by the music and works as a loudspeaker.

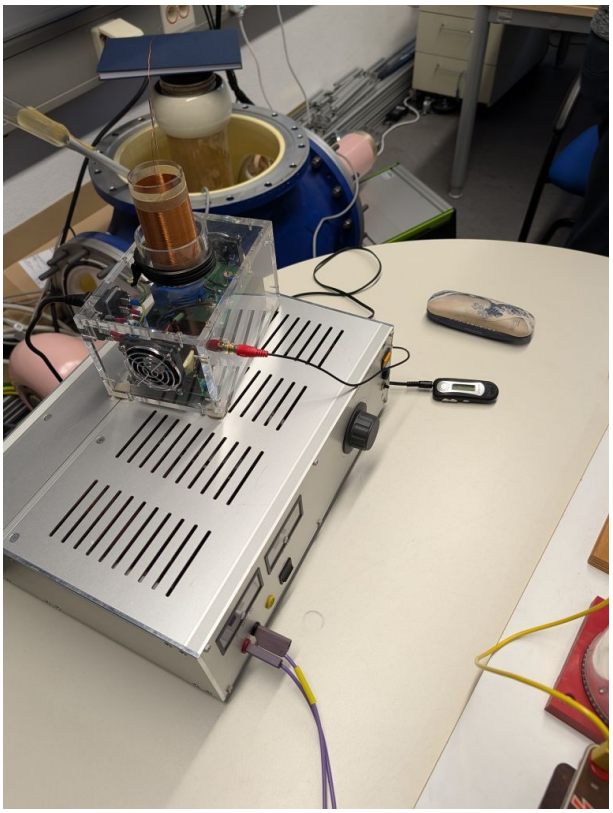


Figure 6. Solid-state Tesla coil on its supply, with the music player connected.

3. Partial discharges in solids

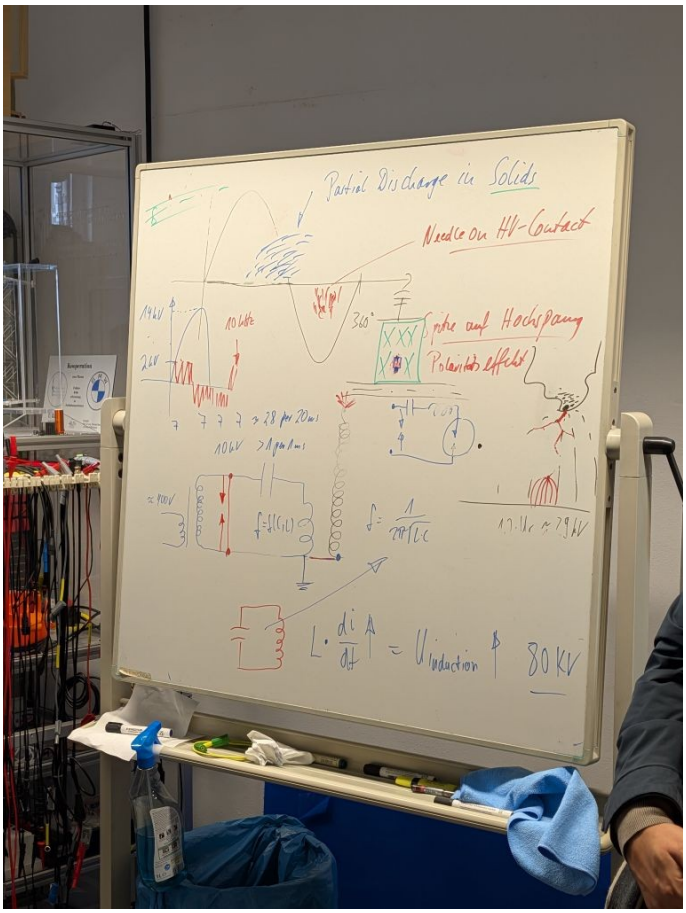


Figure 7. Whiteboard sketch: partial discharges in solids, the Tesla circuit and the spark count.

What is a partial discharge?

A partial discharge (PD) is a local breakdown in a small part of the insulation, for example a gas void or the region around a sharp tip, that does not bridge the two electrodes. Each discharge is small, but repeated discharges break chemical bonds, heat and carbonise the material and erode it until the insulation fails (Dielectric insulation and high-voltage issues, CERN accelerator school).

Needle on the high-voltage contact

The experiment uses a needle on the high-voltage contact ("Spitze auf Hochspannung"), embedded in a solid insulating sample, against a flat earthed electrode. At the needle tip the field is many times higher than the average field across the sample, so discharges start there at a voltage far below the breakdown voltage of the whole sample.

The needle-plane arrangement is the standard laboratory geometry for studying partial discharges and electrical trees in solids (University of Manchester, treeing in epoxy resin).

Where on the sine wave do discharges occur?

As in the spark-gap example, a discharge site fires each time the local voltage has changed by its inception voltage since the last discharge. Discharges therefore appear as groups of pulses on the rising and falling parts of the sine wave, and their number per cycle grows with the applied voltage. For treeing, the repetition

rate is often highest where the voltage changes fastest, and the PD magnitude increases with voltage (Partial discharge in high voltage insulating materials).

Polarity effect

With a needle electrode the discharge pattern differs between the positive and negative half-waves. Charge injected at the tip forms a space-charge cloud that changes the local field; for needle-plane samples, positive polarity generally starts electrical trees at a lower voltage than negative polarity (Chalmers University of Technology thesis).

Electrical treeing

Under continued stress, partial discharges at the needle tip produce a branching network of hollow channels that grows through the solid, like the branches of a tree. When a channel reaches the opposite electrode, the insulation breaks down. Treeing is a major ageing mechanism in polymer-insulated cables (CERN accelerator school).

Measurement

Partial discharges are measured with a coupling capacitor connected in parallel with the sample and a measuring impedance, as sketched on the whiteboard. Each discharge draws a short current pulse from the coupling capacitor, which the measuring instrument records as an apparent charge in pC together with its position on the sine wave. The measuring method is standardised in IEC 60270.

4. Safety

- Keep the distance shown by the lab staff from both Tesla coils; the high-frequency discharge can cause burns.
- Do not touch any part of the set-up while the supply is on. The 14 kV supply transformer is dangerous even when the Tesla coil itself is not sparking.
- Discharges in air produce ozone; keep the room ventilated.
- The PD experiment is energised only by the lab staff, behind the barrier or interlock.

5. Questions for the report

1. Explain why a 2 kV spark gap fires 28 times per 50 Hz cycle on a 14 kV peak supply. How many sparks per cycle would you expect at 10 kV peak?
2. Why must the primary and secondary circuits of the Tesla transformer be tuned to the same frequency?
3. Why does a partial discharge not lead to immediate breakdown, and why is it still dangerous for the insulation?
4. Describe the polarity effect for a needle electrode and its cause.

References

- Wikipedia, Tesla coil: https://en.wikipedia.org/?title=Magnifying_transmitter
- Western Washington University, High-Frequency Tesla Apparatus: https://physics.wwu.edu/high-frequency-tesla-apparatus

- 3B Scientific, Tesla transformer 1000966, instruction sheet: https://www.3bscientific.com/product-manual/1000966_EN.pdf
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- Chalmers University of Technology, electrical treeing thesis: https://publications.lib.chalmers.se/records/fulltext/138671.pdf
- University of Manchester, electrical treeing in a glassy epoxy resin: https://pure.manchester.ac.uk/ws/files/63515200/Treeing_in_a_glassy_resin_CEIDP_Submitted.pdf
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- T. Betz, Advanced High Voltage Technology, lecture notes, h_da. Available from the lab staff.

Standard: IEC 60270 High-voltage test techniques, Partial discharge measurements.